

```
# STEP 1: Install spacy and download English model
!pip install spacy
!python -m spacy download en_core_web_sm

Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cytoolz<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy<=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic<=1.8.1,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.12.3)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging<=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (26.0)
Requirement already satisfied: annotated-types<=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.7.0)
Requirement already satisfied: pydantic-core<2.41.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (2.41.4)
Requirement already satisfied: typing-extensions<=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic<=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (4.15.0)
Requirement already satisfied: typing-inspection<=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.4.2)
Requirement already satisfied: charset-normalizer<=4.3.2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.5)
Requirement already satisfied: idna<=4.3 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.11)
Requirement already satisfied: urllib3<=3.1.2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.5.0)
Requirement already satisfied: certifi<=2024.12.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2025.2.25)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.3)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.1.5)
Requirement already satisfied: smart-open<8.0.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.5.1)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.5.1)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.5.1)
Requirement already satisfied: click<=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (8.3.1)
Requirement already satisfied: shellingham<1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (1.5.4)
Requirement already satisfied: rich<=13.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (13.9.4)
Requirement already satisfied: annotated-doc<=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (0.0.4)
Requirement already satisfied: markdown-it-py<=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich<=13.0->typer-slim<1.0.0,>=0.3.0->spacy) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich<=13.0->typer-slim<1.0.0,>=0.3.0->spacy) (2.19.2)
Requirement already satisfied: mdurl<=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py<=2.2.0->rich<=13.0->typer-slim<1.0.0,>=0.3.0->spacy) (0.1.2)
Collecting en-core-web-sm<3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-any-none-any.whl (12.8 MB)
    12.8/12.8 MB 85.3 MB/s eta 8:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the "Restart kernel" or "Restart runtime" option.
```

```
import spacy

# Load the English model
nlp = spacy.load("en_core_web_sm")
print("Model loaded successfully!")

Model loaded successfully!
```

```
# STEP 2: Prepare dataset (15 sentences across categories)
sentences = [
    "The cat sat on the mat.", # Simple
    "She likes tea and he likes coffee.", # Compound
    "Although it was raining, we went outside.", # Complex
    "Did you finish your homework?", # Interrogative
    "The book was read by the student.", # Passive
    "Breaking news: The stock market crashed today.", # News
    "I do that movie was amazing!", # Social media
    "The algorithm achieved 95% accuracy on the dataset.", # Technical
    "He runs fast.", # Simple
    "I wanted to go, but I stayed home.", # Compound
    "Because she was tired, she went to bed early.", # Complex
    "Who is your best friend?", # Interrogative
    "The homework was completed by John.", # Passive
    "Scientists discovered water on Mars.", # News
    "OMG this food is so good!" # Social media
]
```

```
# Process sentences with spacy
docs = [nlp(sent) for sent in sentences]
```

```
import pandas as pd

# STEP 3: Extract token details
results = []
for doc in docs:
    for token in doc:
        results.append([doc.text, token.text, token.lemma_, token.pos_, token.dep_, token.head.text])

df = pd.DataFrame(results, columns=["Sentence", "Token", "Lemma", "POS", "Dependency", "Head"])
df.head(30) # Show first 30 rows
```

	Sentence	Token	Lemma	POS	Dependency	Head
0	The cat sat on the mat.	The	the	DET	det	cat
1	The cat sat on the mat.	cat	cat	NOUN	nsbj	sat
2	The cat sat on the mat.	sat	sit	VERB	ROOT	sat
3	The cat sat on the mat.	on	on	ADP	prep	sat
4	The cat sat on the mat.	the	the	DET	det	mat
5	The cat sat on the mat.	mat	mat	NOUN	potj	on
6	The cat sat on the mat.	.	.	PUNCT	punct	sat
7	She likes tea and he likes coffee.	She	she	PRON	nsbj	likes
8	She likes tea and he likes coffee.	likes	like	VERB	ROOT	likes
9	She likes tea and he likes coffee.	tea	tea	NOUN	doj	likes
10	She likes tea and he likes coffee.	and	and	CCONJ	cc	likes
11	She likes tea and he likes coffee.	he	he	PRON	nsbj	likes
12	She likes tea and he likes coffee.	likes	like	VERB	conj	likes
13	She likes tea and he likes coffee.	coffee	coffee	NOUN	doj	likes
14	She likes tea and he likes coffee.	.	.	PUNCT	punct	likes
15	Although it was raining, we went outside.	Although	although	SCONJ	mark	raining
16	Although it was raining, we went outside.	it	it	PRON	nsbj	raining
17	Although it was raining, we went outside.	was	be	AUX	aux	raining
18	Although it was raining, we went outside.	raining	rain	VERB	advcl	went
19	Although it was raining, we went outside.	.	.	PUNCT	punct	went
20	Although it was raining, we went outside.	we	we	PRON	nsbj	went
21	Although it was raining, we went outside.	went	go	VERB	ROOT	went
22	Although it was raining, we went outside.	outside	outside	ADV	advmod	went
23	Although it was raining, we went outside.	.	.	PUNCT	punct	went
24	Did you finish your homework?	Did	do	AUX	aux	finish
25	Did you finish your homework?	you	you	PRON	nsbj	finish
26	Did you finish your homework?	finish	finish	VERB	ROOT	finish
27	Did you finish your homework?	your	your	PRON	poss	homework
28	Did you finish your homework?	homework	homework	NOUN	doj	finish
29	Did you finish your homework?	?	?	PUNCT	punct	finish

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# STEP 4: Identify subject, verb, object, modifiers
for doc in docs:
    print(f'Sentence: {doc.text}')
    subject = [token.text for token in doc if token.dep_ == "nsbj"]
    verb = [token.text for token in doc if token.dep_ == "ROOT"]
    obj = [token.text for token in doc if token.dep_ == "doj"]
    modifiers = [token.text for token in doc if token.dep_ in ["amod", "prep"]]

    print(f'Subject: {subject}')
    print(f'Main Verb: {verb}')
    print(f'Object: {obj}')
    print(f'Modifiers: {modifiers}')
```

Modifiers: []

```
Sentence: The algorithm achieved 95% accuracy on the dataset.
Subject: ['algorithm']
Main Verb: ['achieved']
Object: ['accuracy']
Modifiers: ['on']

Sentence: He runs fast.
Subject: ['He']
Main Verb: ['runs']
Object: []
Modifiers: []

Sentence: I wanted to go, but I stayed home.
Subject: ['I', 'I']
Main Verbs: ['wanted']
Object: []
Modifiers: []

Sentence: Because she was tired, she went to bed early.
Subject: ['she', 'she']
Main Verb: ['went']
Object: []
Modifiers: ['to']

Sentence: Who is your best friend?
Subject: ['friend']
Main Verb: ['is']
Object: []
Modifiers: ['best']

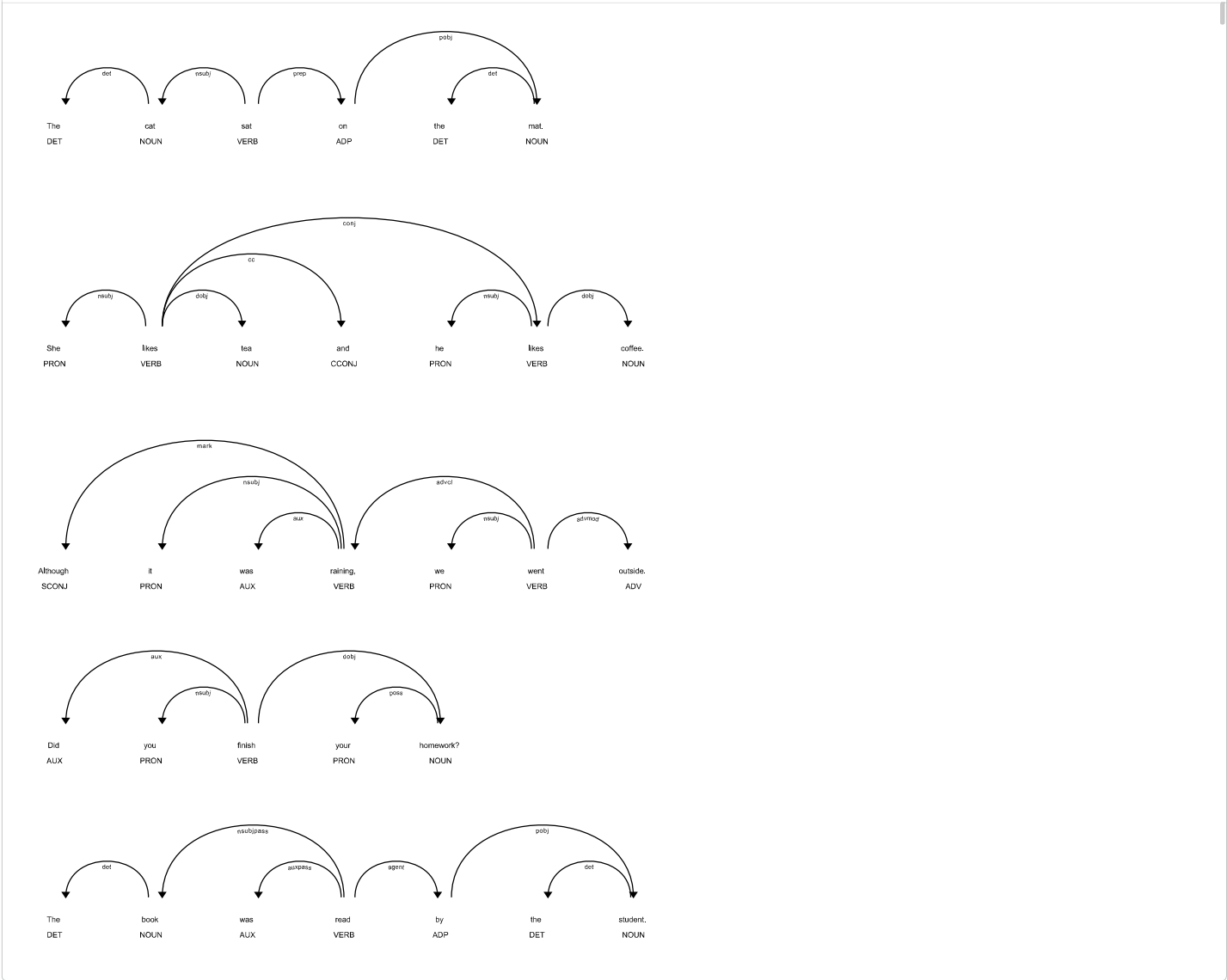
Sentence: The homework was completed by John.
Subject: []
Main Verb: ['completed']
Object: []
Modifiers: []

Sentence: Scientists discovered water on Mars.
Subject: ['Scientists']
Main Verb: ['discovered']
Object: ['water']
Modifiers: ['on']

Sentence: OMG this food is so good!
Subject: ['OMG']
Main Verb: ['is']
Object: []
Modifiers: []
```

```
from spacy import displacy

# STEP 5: Visualize dependency trees for first 5 sentences
for doc in docs[:5]:
    displacy.render(doc, style="dep", jupyter=True)
```



```
# STEP 6: Comparative analysis notes
analysis = """
1. Simple sentences show clear subject-verb-object relations.
2. Compound sentences have multiple ROOT verbs connected by conjunctions.
3. Complex sentences introduce subordinate clauses, making trees deeper.
4. Interrogative sentences shift word order but still show subject-verb relations.
5. Passive voice changes dependency: subject becomes object of 'by' phrase.
6. Modifiers (adjectives, prepositions) expand tree branches.
7. News sentences often contain multiple modifiers and compound structures.
8. Social media sentences may include interjections (LOL, OMG) that appear as discourse elements.
"""
print(analysis)
```

```
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